BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: 112101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Rate of change of angular momentum is equal to.

(A) Force
(B) Power
(C) Torque
(D) Work

Q2. Terminal velocity of spherical object is given by.

(A) mg/4πη
(B) mg/6πη
(C) mg/6πη
(D) mg/9πη

Q3. A pipe has cross section area $0.0001m^2$ in which water flows at 10m/s. The volume flow rate is.

(A) $0.00001m^3/s$
(B) $0.001m^3/s$
(C) $1m^3/s$
(D) $0.0001m^3/s$

Q4. Time period of the same pendulum at Karachi and Muree are related as.

(A) TK=TM
(B) TK > TM
(C) TK < TM
(D) 2TK = 3TM

Q5. The restoring force of SHM is maximum when particle.

(A) Displacement is maximum
(B) Half way between them
(C) Crossing mean position
(D) At rest

Q6. Wave generator produces 500 pulses in 10sec. Find the frequency of the pulses produced in second.

(A) 5000
(B) 500
(C) 50
(D) 1/50

Q7. Which of the following factor has no effect on the speed of sound in gas?

(A) Humidity
(B) Pressure
(C) Temperature
(D) Density

Q8. Coloured fringes observed in a soap bubbles are example of.

(A) Diffraction
(B) Interference
(C) Polarization
(D) Reflection

Q9. The S.I unit of molar specific heat or molar specific heat capacity.

(A) $Jmole^{-1}$
(B) J/kg
(C) $J\ mole^{-1}\ K^{-1}$
(D) $Kg\ J^{-1}$

Q10. Dimension of torque is.

(A) [MLT]
(B) $[M^2L^2T]$
(C) $[ML^2T^{-2}]$
(D) $[ML^2T^2]$

Q11. The sum of magnitudes of two forces is 16N. If the resultant force is 8N and its direction is perpendicular to minimum force then the forces are.

(A) 6N and 10N
(B) 8N and 8N
(C) 4N and 12N
(D) 2N and 14N

Q12. Time rate of change of momentum gives.

(A) Force
(B) Impulse
(C) Acceleration
(D) Power

Q13. Area between velocity time graph is numerically equal to.

(A) Velocity
(B) Displacement
(C) Acceleration
(D) Power

Q14. Dimension of work is.

(A) $[MLT^2]$
(B) $[MLT^{-2}]$
(C) $[ML^2T^{-2}]$
(D) $[ML^2T^2]$

Q15. Torque has zero value, if the angle between r and F is.

(A) 0°
(B) 30°
(C) 45°
(D) 90°

Q16. Expression for escape velocity is given by.

(A) √(2GRe)
(B) √(GMe)
(C) √(2gRe)
(D) √(gRe)

Q17. The atmosphere is held to the earth by.

(A) Wind
(B) Gravity
(C) Cloud
(D) Rotation of earth

Q18. Moment of inertia of rod or meter stick is given by.

(A) 1/6 Ml^2
(B) 1/9 Ml^2
(C) 1/12 Ml^2
(D) 1/4 Ml^2

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 40

QI. Attempt any Ten Parts out of the following. Each Part carries equal marks.

(i) Why including more digits in answers, does not make it more accurate?

(ii) A central force is one that is always directed toward the same point-can a central force give rise to a torque about that point?

(iii) Why we can hit a long sixer in a cricket match rather than if we toss a ball for our selves?

(iv) What is the angle for which the maximum height reached and corresponding range are equal?

(v) A bucket is taken to the bottom of a well, does the bucket possess any P.E? Explain

(vi) Why is the acceleration of a body moving uniformly in a circle, directed towards the centre?

(vii) A ball is just supported by a string without breaking. If it is set swinging, it breaks. Why?

(viii) You can squirt water over a greater distance by placing your thumb over the end of a garden hose, than by leaving it completely uncovered. Explain how it works?

(ix) What will be the frequency of simple pendulum if its length is "0.5m".

(x) Why in S.H.M the acceleration is zero when the velocity is greatest?

(xi) Clearly explain the difference between Longitudinal and transverse waves.

(xii) Why it is not possible to see the interference where the Light beams from the head lamps of a car over lap?

(xiii) Can specific heat of a gas be zero or infinity? Can specific heat be negative?

SECTION "C"

Maximum Marks: 27

Note: Attempt any Three questions of the following. Each question carries equal Marks.

QIII.

(a) [5] What is meant by dimensions of Physical quantities? What are limitations and applications of dimensional analysis?

(b) [4] A football leaves the foot of a punter at an angle of 54° (Positive x-direction) at a speed of 21m/s. Determine the horizontal and vertical components of velocity.

QIV.

(a) [5] What is linear momentum? Derive and state Newton's second law in terms of linear momentum

(b) [4] A man pulls a trolley through a distance of 50m by applying a force of 100N which makes an angle of 30° with horizontal. Calculate the work done by the man.

QV.

(a) [5] What is terminal velocity? Derive mathematical relation for terminal velocity by using stokes law.

(b) [4] A circular drum of radius 40cm is initially rotating at 400 rev/min. It is brought to stop after making 50 revolutions. What is it's angular acceleration and stopping time?

QVI. Write down notes on the two of the following. a) S.H.M b) Second Law of thermodynamics c) Doppler's effect.